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$$7(1+i)z = (1-i)(12-z^2)$$

$$\Rightarrow 7\frac{1+i}{1-i}z = 12 - z^2$$

$$\Rightarrow 7iz = 12 - z^2$$

$$\Rightarrow z^2 + 7iz - 12 = 0$$

$$\Delta = (7i)^2 - 4(1)(-12) = -49 + 48 = -1$$

$$\Rightarrow z_1 = \frac{-7i+i}{2} = -3i, z_2 = \frac{-7i-i}{2} = -4i$$

$$\Rightarrow z_1 = -3i, z_2 = -4i$$

References